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SELF-SUPERVISED LEARNING FOR INSAR PHASE AND COHERENCE ESTIMATION

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ABSTRACT

This paper focuses on the estimation of interferometric SAR parameters, a step that precedes the entire interferometric processing chain to produce derived information such as digital elevation models and ground displacement. Deep learning, especially convolutional neural networks (CNN), has revolutionized image denoising and has recently received considerable attention. However, traditional supervised approaches require labeled images for training, which are generally unavailable or inaccurate, especially in remote sensing applications. To overcome this limitation, semi- and self-supervised denoising approaches have recently been proposed. These can learn from exclusively noisy samples, which can be obtained from pairs of noisy images or from noisy values within the same image. In this paper, we build on the foundation of these self-supervised learning methods, in particular, we borrow concepts from the Noise2Void and Noise2Self approaches, which have already shown excellent performance in various image denoising tasks. We extend this method to address the challenges specific to InSAR phase and coherence estimation, where the complex-valued nature of SAR interferograms poses unique processing considerations.

Index Terms— synthetic aperture radar, SAR, InSAR, deep learning, self-supervised learning

1. INTRODUCTION

Denoising is crucial in improving the performance and accuracy of various image processing tasks. It aims to recover the true underlying pixel values from noisy observations. In the field of synthetic aperture radar (SAR) remote sensing, denoising techniques are widely used for despeckling backscatter data and improving the quality of interferometric SAR (InSAR) derived products, such as digital elevation models (DEM) and ground displacement estimates. Deep learning has revolutionized the field of image denoising and is growing rapidly. The first widespread solution using convolutional neural networks (CNN) has been proposed in [1]. It proposes denoising in a supervised fashion by estimating the image noise, which is eventually subtracted from the input image. Supervised methods generally require labeled training data and thus the noiseless image itself, which is usually unavailable since all sensor-acquired images are usually affected by noise. Therefore, supervised training resorts to simulate noise-free images and superimpose the noise to provide labeled training data. If the simulation is performed correctly, i.e., if it properly considers the physics involved in the real image generation process, the resulting high-quality training data makes the performance of supervised denoising methods excel. Still, performance drops quickly when the statistical distribution of real data differs from the noise model used for simulation. Therefore, finding an alternative to supervised learning is crucial to improve algorithms' accuracy and robustness. In the last couple of years, several semi- and self-supervised approaches to image denoising have been proposed to cope with this issue. Among the first and most popular ones, it is worth mentioning the Noise2Noise [2], Noise2Void [3], and Noise2Self [4] approaches. The key idea behind these algorithms, first proposed in the Noise2Noise paper, is that it is possible to train a deep learning model to denoise an image by using only pairs of noisy images as input without the need for any clean (noise-free) images. This reasoning assumes that, contrarily to the underlying noise-free signal, noise is spatially uncorrelated. Some of these have already found application in the SAR despeckling domain. Following the Noise2Noise methodologies, the SAR2SAR [5] and the Merlin [6] approaches have been proposed. Similarly, Speckle2Void [7] has been proposed as an adaptation of the Noise2Void approach to SAR speckle statistics. Contrarily to the despeckling field, InSAR parameter estimation still needs to be addressed with self-supervised methodologies. With the InSAR signal being complex-valued, self-supervised approaches formulated for real-valued data are not directly applicable, and different processing design choices should be made. Indeed, the real-valued interferometric phase cannot be processed as a real-valued image, given its wrapped nature. Possible solutions rely on separating the complex-valued interferogram into its real and imaginary parts and processing them separately, as done in [8, 9, 10]. Alternatively, one could resort to the use of complex-valued CNNs, which allow the use of a complex input but are not yet investigated together with self-supervised approaches. This paper presents a self-supervised approach based on blind spot networks, as presented in the Noise2Void and Noise2Self approaches, and provides a solution for the processing of complex-valued signals, proving the feasibility of such a method for the interferometric SAR signal.

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The paper is organized as follows: In section 2, we present the InSAR noise model and the adopted self-supervised methodology. In section 3, we present the experiments and the results. Finally, in section 4, we summarize the presented work and comment on possible future developments.

2. METHODOLOGY

After coregistration of the *secondary* single-look-complex (SLC) image u_s on the grid of the *primary* image u_p used as reference, an interferogram v can be obtained by multiplying the primary SLC by the complex conjugate of the secondary

$$v = u_p \cdot u_s^* = |u_p| \cdot |u_s| \cdot \exp(j\phi) \tag{1}$$

where the symbol $*$ indicates the complex conjugate operator and, ϕ is the interferometric phase, which, being φ the phase of the SLC, can be written as

$$\phi = \varphi_p - \varphi_s \tag{2}$$

The interferometric phase encodes information about the possible differences between the two SLCs due to the slightly different acquisition geometry and time, thus providing information about topography and temporal ground displacement.

Another very informative quantity for interferometric applications is the normalized complex correlation between the two SLCs, often referred to as *complex coherence* and defined as follows

$$\hat{\gamma} = \frac{E[u_p \cdot u_s^*]}{\sqrt{E[|u_p|^2] \cdot E[|u_s|^2]}} = \hat{\rho} \cdot \exp\left(j\hat{\phi}\right) \tag{3}$$

where E denotes the expectation value that can be replaced by a spatial mean in a maximum likelihood framework. The quantity $\hat{\phi}$ indicates an estimate of the interferometric phase, and the quantity $\hat{\rho}$ is an estimate of the *coherence*, which indicates the level of correlation (provided in the interval $[0, 1]$) between the two SLCs and thus also information about the local phase noise. In addition, the coherence has been shown to be an excellent proxy for land cover inference and is therefore an additional desired output quantity in interferometric processing.

In the present investigation, we consider a proxy for the noisy complex coherence similar to that proposed in [10]

$$\gamma = \frac{u_p \cdot u_s^*}{\sqrt{E[|u_p|^2] \cdot E[|u_s|^2]}} \tag{4}$$

and split it into its real and imaginary parts in order to process them independently (for simplicity, we disregard at this stage of the investigation the possible correlation between the real and imaginary parts of the interferometric signal). Therefore, we consider two features as input to our network, a and b, as the real and imaginary parts of γ, respectively, i.e.

$$a = \Re\{\gamma\} \qquad\qquad b = \Im\{\gamma\}$$

We propose a self-supervised training approach that combines concepts from the Noise2Void [3] and Noise2Self [4] methods. Following the self-supervised paradigm, we aim to train a neural network without relying on manually labeled data, but instead using a given transformation on the input data. Specifically, we use *blind spot* masks to occlude parts of the input image at random locations. By applying these masks, we remove the information about the input values at these specific locations from the network during training. Therefore, the network is trained to determine the relationship between the input and output data without knowing the exact values of the input pixels covered by the blind spot masks, and is forced to infer the missing pixel values by relying on the neighboring pixels that are still visible in the image. An analogy to illustrate this concept is the use of a boxcar filter, where the value of a pixel is calculated as the average of its neighboring pixels within a given window. Similarly, our network will depend on neighboring pixels, but instead of using a fixed operation like the averaging of the boxcar filter, it will learn the appropriate operation from the available data. During training, the blind spot masks, as well as the input data, are varied at each training iteration. We use an MSE loss function between the noisy input and the estimated values. While we estimate a full image without blind spots, we restrict the computation of the loss function to the pixels in the blind spots. Thus, the final estimate of the interferometric parameters is computed as follows

$$\hat{\phi} = \arctan\left(\frac{\hat{b}}{\hat{a}}\right) \qquad\qquad \hat{\rho} = \sqrt{\hat{a}^2 + \hat{b}^2}$$

This strategy is agnostic to the specific type of architecture used. Based on previous experience, we adopt a U-Net-based architecture similar to the one used in [10], but without the residual connection between input and output.

3. EXPERIMENTS AND RESULTS

We performed experiments on high-resolution interferometric SAR images from the TanDEM-X mission. Specifically, we used 10^4 square patches of 256×256 pixels and trained the network on 500 epochs with a batch size of 8 and an Adam optimizer with a fixed learning rate of 10^{-4}. For this preliminary investigation, we compare our results with those obtained exclusively for state-of-the-art methods that do not require clean data references, such as the OC-InSAR-BM3D method. As a low resolution reference, we also show results obtained with a 3×3 boxcar filter. Figure 1 and Figure 2 show the comparison between these three estimators for two test patches,

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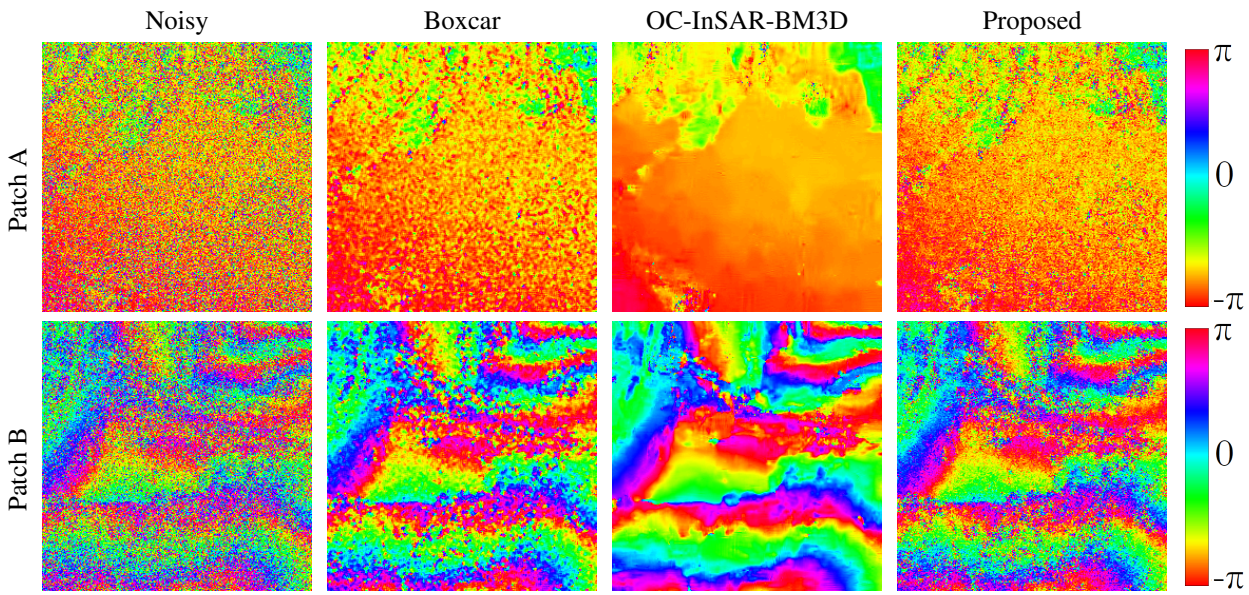

Fig. 1. Interferometric phase images of the considered TanDEM-X interferogram patches estimated using the reference methods and our proposed one.

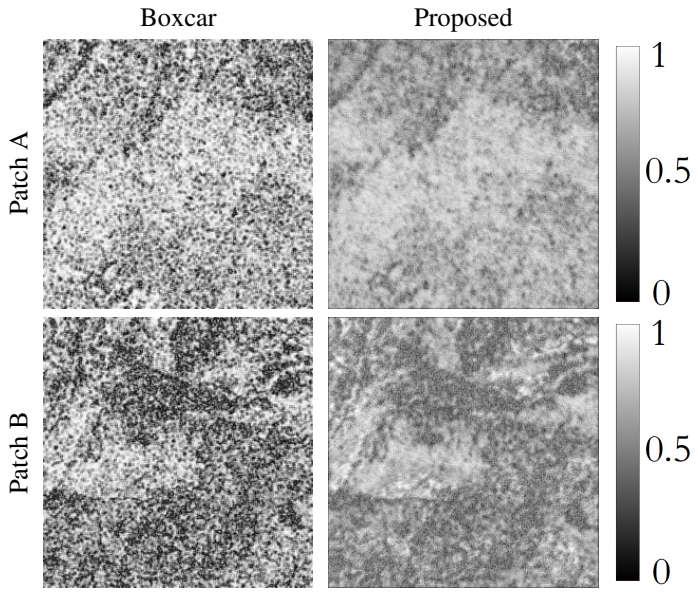

Fig. 2. Interferometric coherence images of the considered TanDEM-X interferogram patches estimated using the reference methods and our proposed one.

marked with the letters A and B, for the interferometric phase and coherence.

From Figure 1 we can see that the algorithm has good denoising capabilities, although it does not reach the level of performance achieved by OC-InSAR-BM3D. The algorithm preserves many details of the input image, which is a plus, indicating that the denoising process does not result in significant loss of important features or information present in the original image. Patch A, which has constant phase values, would typically require higher denoising capabilities. In this respect, on the one hand, the denoising performance of the proposed algorithm may be somewhat lacking, suggesting potential for further improvement to effectively reduce noise in regions with uniform phase behavior. On the other hand,

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phase jumps are better preserved compared to OC-InSAR-BM3D, which generally shows a smoother behavior. Patch B, which has some topography and interferometric fringes, shows a better result. In this case, the algorithm shows good denoising ability, effectively reducing noise while preserving the underlying details of the topography. This result indicates that the algorithm is capable of handling more complex features and preserving the necessary details even in regions with distinct interferometric features. From Figure 2, we can see that the results of the proposed algorithm for the estimated coherence for both patches closely follow the results of the boxcar filter and also show a higher denoising capability. However, it is also clear that there is a bias in the results. The observed trade-off between denoising capability and smoothness of the result, as well as the coherence bias, suggests that the approach has potential, but also highlights the need for further investigation.

4. SUMMARY & CONCLUSION

This research demonstrates the potential of self-supervised learning methods for InSAR phase and coherence estimation. Without the need for reference data, the network is able to learn the features present in the interferometric phase and coherence. While the self-supervised method presented in this paper shows encouraging performance, further research is needed to address some limitations and challenges, such as the trade-off between denoising capability and detail preservation and unbiased coherence estimation. In addition, the handling of complex-valued SAR data warrants further investigation, especially in exploring the use of different losses and different self-supervised strategies.

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